

ADVANCED PERSISTENT THREATS

APT GROUP ANALYSIS

TARGET: RUSSIA

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Abstract

This article examines **Russia** as the focus of our analysis of Advanced Persistent Threat (APT) groups. We will explore the campaigns attributed to Russian threat actors, their operational objectives, and the intelligence structures believed to support many of these cyber operations.

Keywords: Cybersecurity, Cybercrime, APT, Threat, Russia

INTRODUCTION

Russia is widely regarded as one of the world's leading cyber powers. Throughout modern history, its extensive experience in **Signals Intelligence (SIGINT)** and intelligence operations has led researchers to attribute a large number of Advanced Persistent Threat groups to Russian actors.

At the time this article was written, **ThaiCERT** listed **49 APT groups** attributed to Russia, making it one of the countries with the highest number of publicly identified threat actors.

Covering every aspect of Russian cyber operations in a single article would be impossible. Russian intelligence activities extend well beyond cyberattacks and include traditional **Human Intelligence (HUMINT)**, counterintelligence, influence operations, and decades of institutional evolution following the dissolution of the **KGB**.

Many former members of Soviet intelligence services later assumed influential roles within politics, security agencies, and organized crime, contributing to the complex ecosystem surrounding modern Russian intelligence operations.

FROM INTELLIGENCE SERVICES...

Given this background, it is reasonable to assume that at least some Russian-attributed APT campaigns receive some form of state support.

This is a significant observation because Russia possesses substantial technological, military, and intelligence resources. During **2020**, ThaiCERT documented activity from **17 of the 49** Russian-attributed APT groups.

Among these groups:

- At least **seven** were explicitly classified by ThaiCERT as **state-sponsored**.
- At least **ten** were believed to have operational links with Russian intelligence agencies.

The intelligence organizations most frequently associated with these operations include:

FSB (Federal Security Service)

Responsible for:

- Domestic intelligence
 - Counterintelligence
 - Internal security
 - Electronic surveillance
-

SVR (Foreign Intelligence Service)

Specialized primarily in:

- Human Intelligence (HUMINT)
 - Signals Intelligence (SIGINT)
 - Foreign intelligence collection
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FSO (Federal Protective Service)

Responsible for protecting senior government officials while also maintaining significant SIGINT capabilities.

GRU (Main Directorate of the General Staff)

Perhaps the intelligence organization most frequently linked to Russian APT groups.

The GRU possesses capabilities across virtually every intelligence discipline, including:

- Open-Source Intelligence (OSINT)
- Signals Intelligence (SIGINT)
- Human Intelligence (HUMINT)
- Imagery Intelligence (IMINT)

Because of these capabilities, many public reports associate GRU units with some of Russia's best-known cyber operations.

...TO RUSSIAN APT GROUPS

Among the numerous Russian-attributed groups, several stand out because of their scale, sophistication, and operational history.

APT28 AND APT29

According to ThaiCERT, **APT28** has operated continuously from at least **2011 through 2020**.

The group is commonly believed to consist of **GRU Units 26165 and 74455**, while **APT29** is often considered to operate alongside or in coordination with elements of the Russian intelligence community.

Whether APT29 represents an independent organization remains uncertain.

Possible explanations include:

- A separate GRU unit.
- Another intelligence agency, such as the FSB.
- An independent operational team.
- A combination of several organizations.

As with most APT attributions, definitive evidence remains unavailable.

Targeted Sectors

APT28 and APT29 have targeted an exceptionally broad range of industries, including:

- Automotive
- Aerospace
- Chemical manufacturing
- Construction
- Defense
- Education
- Diplomatic missions
- Engineering
- Financial services
- Government institutions
- Healthcare
- Industrial organizations
- Information Technology
- Media
- Non-governmental organizations (NGOs)
- Oil and gas
- Think tanks
- Intelligence agencies

Their operational scope demonstrates that intelligence gathering—not financial crime—is their primary objective.

COUNTRIES TARGETED BY APT28 AND APT29

The operational reach of APT28 and APT29 extends across virtually every continent. Publicly documented campaigns have targeted organizations located in:

- Belgium
- Brazil
- Bulgaria
- Canada
- China
- France
- Germany
- Italy
- Malaysia
- Montenegro
- The Netherlands
- Norway
- Poland
- Romania
- Serbia
- Spain

- Turkey
- Ukraine
- The United Kingdom
- The United States

The diversity of these targets illustrates the strategic nature of Russian cyber operations, which are designed to support foreign intelligence objectives rather than focus on a single geographic region.

COVID-19 ESPIONAGE CAMPAIGNS

During the COVID-19 pandemic, several Western governments publicly accused Russian APT groups of attempting to compromise organizations involved in vaccine research and pharmaceutical development.

According to those reports, the attacks focused on:

- Universities conducting medical research.
- Pharmaceutical companies.
- Public health institutions.
- Laboratories involved in vaccine development.

The alleged objective was to obtain scientific research and accelerate domestic vaccine programs through cyber espionage.

As in many other attribution cases, Russia denied any involvement in these operations.

TURLA

Among all Russian-attributed Advanced Persistent Threat groups, **Turla** is considered one of the most technically sophisticated.

Researchers believe the group has been active since at least **2004**, making it one of the longest-running cyber espionage operations ever documented.

Turla specializes in:

- Long-term cyber espionage.
- Intelligence collection.
- Persistence inside highly protected networks.
- Custom malware development.

Unlike financially motivated cybercriminal organizations, Turla primarily targets:

- Government agencies.
- Diplomatic missions.
- Military organizations.
- Research institutions.
- Critical infrastructure.

Its malware frequently incorporates multiple persistence mechanisms, encrypted communications, modular architectures, and advanced anti-forensic techniques designed to avoid detection for extended periods.

GAMAREDON

Another Russian-attributed group frequently discussed by cybersecurity researchers is **Gamaredon**.

Unlike Turla, which often demonstrates exceptional operational discipline, Gamaredon conducts large-scale campaigns against:

- Ukrainian government agencies.
- Military organizations.
- Critical infrastructure.
- Law enforcement.

The group's primary objective appears to be continuous intelligence collection related to the ongoing conflict between Russia and Ukraine.

Gamaredon is generally associated with sustained phishing campaigns that repeatedly target the same organizations until access is obtained.

DOPPEL SPIDER

Although many Russian groups focus on cyber espionage, others are associated primarily with financially motivated cybercrime.

One example is **Doppel Spider**, believed to be responsible for the **Dridex** banking Trojan.

Dridex has been used extensively to:

- Steal banking credentials.
- Conduct financial fraud.
- Deploy ransomware.
- Establish persistent access to corporate networks.

Unlike intelligence-oriented APT groups, Doppel Spider's activities are generally motivated by financial profit.

Nevertheless, cybersecurity researchers have long debated the extent to which organized cybercriminal groups may operate with varying degrees of tolerance—or indirect support—from national authorities.

THE GRU, NOTPETYA, AND PUBLIC ATTRIBUTION

One of the most significant cyber incidents ever attributed to Russia was the **NotPetya** attack.

Although initially disguised as ransomware, NotPetya rapidly spread worldwide, causing billions of dollars in damages to multinational companies, logistics providers, hospitals, and government institutions.

Several Western governments—including the United States, the United Kingdom, Canada, and Australia—publicly attributed the operation to officers of the **GRU**.

The attack demonstrated how malware originally intended for a regional geopolitical objective could unintentionally affect organizations across the globe.

NotPetya remains one of the clearest examples of the potentially uncontrollable consequences of offensive cyber operations.

CONCLUSION

Russia continues to occupy a central position within the global Advanced Persistent Threat landscape.

The large number of publicly attributed groups, combined with their technical sophistication and operational persistence, reflects decades of investment in intelligence capabilities.

Although definitive attribution remains extremely difficult, consistent behavioral patterns, malware analysis, infrastructure reuse, and intelligence reporting have enabled researchers to associate numerous campaigns with Russian threat actors.

Understanding these groups is essential not only for cybersecurity professionals but also for policymakers and organizations responsible for protecting critical infrastructure from state-sponsored cyber threats.

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